APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: **Simple Electric Circuit Experiment**

SUBJECT: General Science (SCI101)

CLASS & SECTION: Class 6 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: **92 / 100 (Grade A)**

TEACHER DIRECTIVE & PROBLEM FOCUS:

Describe switch, battery, and bulb connectivity with circuit diagram and electrical conductivity rules.

1. Executive Overview & Problem Context

An electric circuit is a continuous, unbroken conductive loop through which electric charges (electrons) migrate under the influence of potential difference. This laboratory experiment investigates closed, open, and short circuit topologies, component functions, directionality of electric current, and empirical conductivity testing across common materials.

2. Core Analysis, Solutions & Systematic Working

- **Circuit Components & Functional Topology:**

  The experimental setup incorporates a 9V DC chemical cell, insulated copper connection leads, a single-pole single-throw (SPST) toggle knife switch, and a 6V miniature incandescent tungsten filament bulb. A closed circuit provides a continuous path from the positive terminal (+) through the load back to the negative terminal (-).

- **Electron Flow vs Conventional Current Direction:**

  Historically, Benjamin Franklin postulated that positive fluid moves from higher to lower potential (conventional current: positive to negative). Modern solid-state physics demonstrates that charge carriers in metals are negatively charged electrons traveling from negative to positive terminals.

- **Conductivity Testing & Material Characterization:**

  By introducing an open test gap in the circuit, various test specimens were examined. Materials rich in free conduction band electrons (copper, aluminum, iron, graphite pencil lead) closed the circuit, lighting the bulb. Materials with tightly bound valence electrons (rubber eraser, dry pine wood, glass rod, plastic ruler) acted as insulators.

- **Ohmic Resistance & Filament Incandescence:**

  When electric current passes through the high-resistance tungsten filament (R "H 15 ohms), Joule heating occurs ($P = I^2R$). The filament temperature exceeds 2,500°C, emitting visible electromagnetic radiation (luminescence) without burning due to the inert argon atmosphere inside the glass envelope.

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3. Detailed Data Matrix & Systematic Findings

Tested Specimen	Material Class	Bulb State	Electrical Property
Copper Strips	Metal	Bright Glow	Superb Conductor (Low R)
Graphite Pencil Core	Carbon Allotrope	Moderate Glow	Non-metal Conductor
Vulcanized Rubber	Polymer	No Light	Dielectric Insulator
Saline Solution (NaCl)	Electrolyte	Steady Glow	Ionic Conductor (Na+, Cl-)
Borosilicate Glass	Silicate	No Light	Complete Insulator

4. Student Conclusions & Reflective Synthesis

A reliable circuit requires a continuous conductive path, an energy potential source (battery), and a consuming load. Switches provide safe manual control over current flow. Fused safety circuits prevent catastrophic thermal runaway caused by unintended short circuits.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Problem Solving & Structured Working: 18 / 20

Analytical Depth & Research Methodology: 28 / 30

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Outstanding analytical depth & presentation! Schematics and circuit calculations are impeccably illustrated."

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